

2026 nash 중간고사 실제 문제들

$$\frac{dJ}{d\theta} = (\hat{y} - y)^T \quad \text{자주 써야 함.}$$

1. $X = n \times l$ matrix

(i) $\frac{d(X^T X)}{dX} =$

(ii) $\frac{d(\sum_{i=1}^n \sum_{j=1}^l X_{ij} \tilde{y}_j)}{dX} =$

(iii) $\longrightarrow =$

2. skip-gram

(i)

(ii) naive skip-gram limitation

(iii)

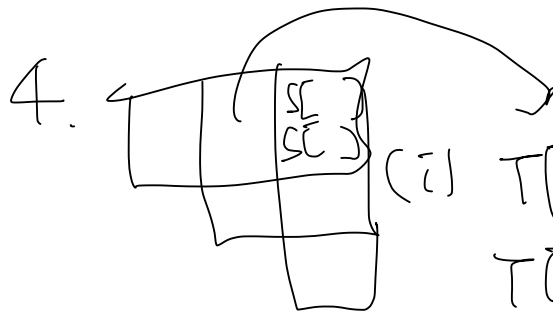
3. Neural Network

여론 편미분 하는 거

i) $\frac{dJ}{dW}$

ii) $\frac{d}{d}$

iii)



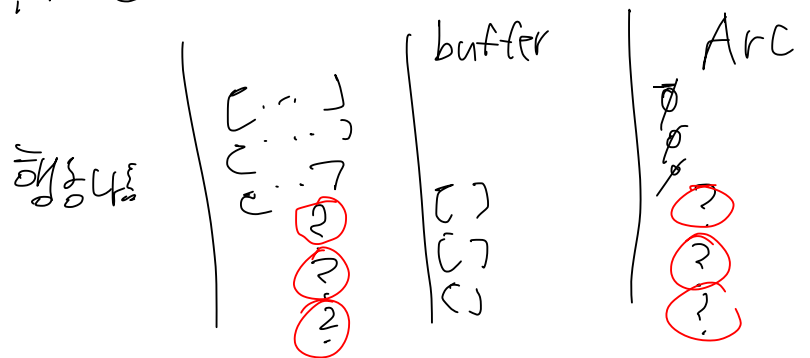
(i) $T[1,2] = \{(WN, 0.02)\}$

$T[2,3] = \{(NP, 0.02)\}$

$T[1,3] = \{(S, 0.03), (S, 0.002)\}$

(ii) Best tree 그려라

5. Arc-standard



6. RNN i) $\frac{\delta J}{\delta h^t}$

ii) $\frac{\delta J}{\delta W}$

iii) $\frac{\delta h^t}{\delta h^{t-1}} ?$

iv) sparsity 때문에 update 어떻게 할지

7. LSTM

i) $\frac{dh^{t+1}}{dh^t}$

ii) Why vanishing gradient 해설?